

WEEK-9- Implementation of Elastic Load Balancer with Auto Scaling using AWS EC2 Instances.

Elastic Load Balancer (ALB)

Step 1: Launch Two EC2 Instances

1.1 Launch EC2 Instance #1

1. Go to **AWS EC2 Console** → **Launch Instance**.
2. **Name**: webserver-1
3. Choose an **AMI** (Amazon Linux 2).
4. Select **Instance Type** (e.g., t2.micro).
5. Add **Storage** (default is fine).
6. Configure **Security Group**:
 - Allow **SSH (22)** from your IP
 - Allow **HTTP (80)** from anywhere (0.0.0.0/0)
7. **Launch** → Select/Create Key Pair → Launch.

Run these commands after connecting VM

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "This is Server 1" > /var/www/html/index.html
```


✓ Now you create second **EC2 instances** running Apache, with message **This is Servier 2** each serving a simple web page.

Repeat it for second instance

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
```

```
echo "This is Server 2" > /var/www/html/index.html
```

The screenshot displays the AWS Management Console interface for an EC2 instance. The browser tabs at the top include 'Launch AWS Academy', 'Instance details | EC2', 'Connect to instance', and others. The URL bar shows the console path for the instance 'i-09b7ac089eaa8f5da' in the 'us-east-1' region.

The left-hand navigation pane shows the 'EC2' section expanded, with options like 'Dashboard', 'Instances', 'Images', 'Elastic Block Store', and 'Network & Security'.

The main content area is titled 'Instance summary for i-09b7ac089eaa8f5da (webserver-2)'. It provides a comprehensive overview of the instance's configuration and status:

- Instance ID:** i-09b7ac089eaa8f5da
- IPV6 address:** -
- Hostname type:** IP name: ip-172-31-31-119.ec2.internal
- Answer private resource DNS name:** IPv4 (A)
- Auto-assigned IP address:** 44.222.209.8 [Public IP]
- IAM role:** -
- IMDSv2:** Required
- Operator:** -
- Public IPv4 address:** 44.222.209.8
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-172-31-31-119.ec2.internal
- Instance type:** t2.micro
- VPC ID:** vpc-0f697a18db9da9c1f
- Subnet ID:** subnet-0566e2767d38d3fe1
- Instance ARN:** arn:aws:ec2:us-east-1:447368333187:instance/i-09b7ac089eaa8f5da
- Private IPv4 addresses:** 172.31.31.119
- Public DNS:** ec2-44-222-209-8.compute-1.amazonaws.com
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations.
- Auto Scaling Group name:** -
- Managed:** false

Below the summary, there are tabs for 'Details', 'Status and alarms', 'Monitoring', 'Security', 'Networking', 'Storage', and 'Tags'. The 'Details' tab is currently selected, showing further instance details.

The bottom of the screen shows a Windows taskbar with the search bar, task view button, and several application icons. The system tray indicates the date and time as 10:49 on 24-03-2026, along with weather information (28°C Sunny).

```
root@ip-172-31-31-119:/home/ec2-user
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
Installing : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 9/13
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13
Installing : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64
httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

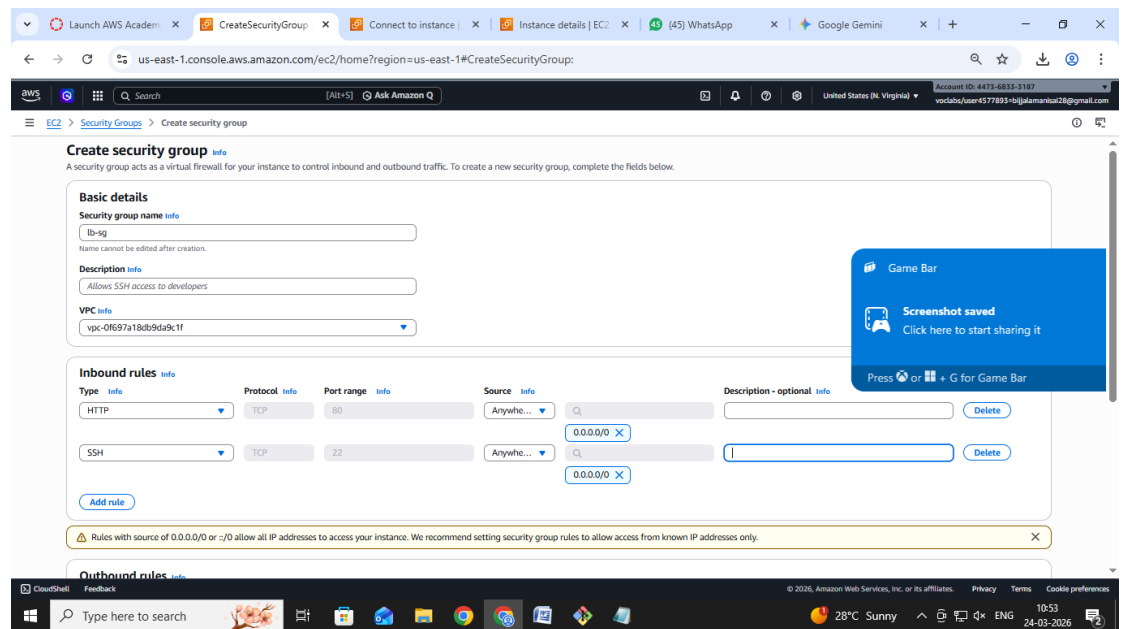
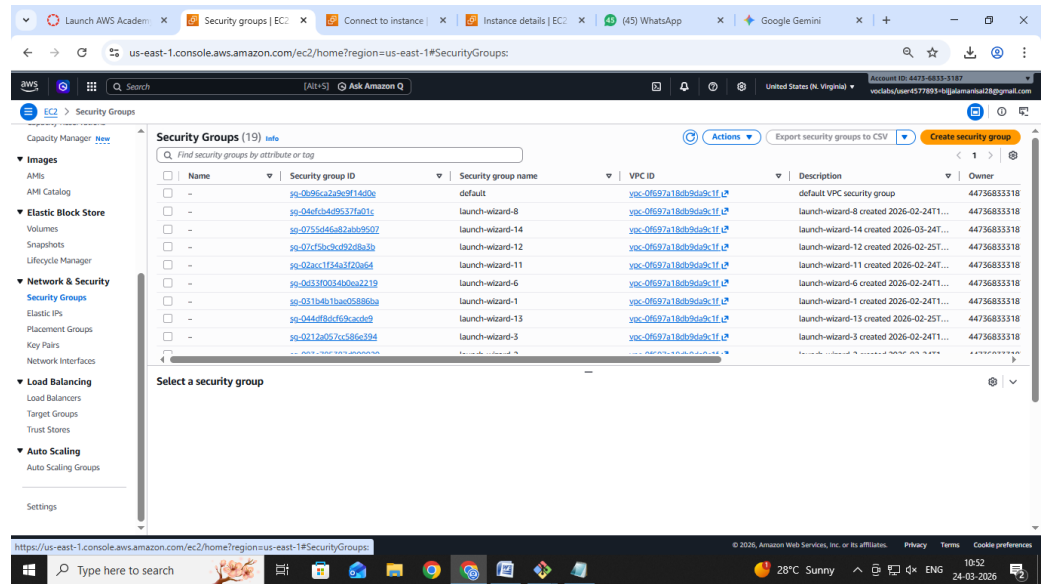
Complete!
[root@ip-172-31-31-119 ec2-user]# systemctl start httpd
[root@ip-172-31-31-119 ec2-user]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service - /usr
/lib/systemd/system/httpd.service.
[root@ip-172-31-31-119 ec2-user]# echo "This is Server 2" > /var/www/html/index
.html
[root@ip-172-31-31-119 ec2-user]#
```

Step 2: Create a Load Balancer

AWS provides **Elastic Load Balancer (ELB)**. We will create an **Application Load Balancer (ALB)**.

2.1 Configure Security Group for Load Balancer

- Go to EC2 → Security Groups → Create Security Group
 - Name: lb-sg
 - Allow **HTTP (80)** from anywhere
 - Optional: HTTPS (443) if using SSL



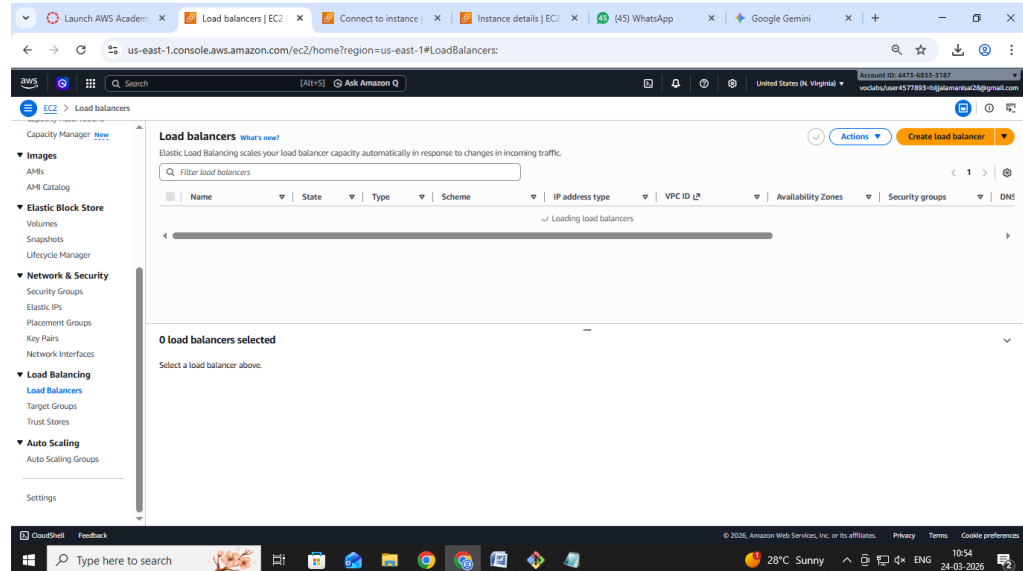
2.2 Create the Load Balancer

1. Go to EC2 → Load Balancers → Create Load Balancer → Application Load Balancer
2. **Basic Configuration:**

- Name: my-alb
- Scheme: Internet-facing
- IP address type: IPv4

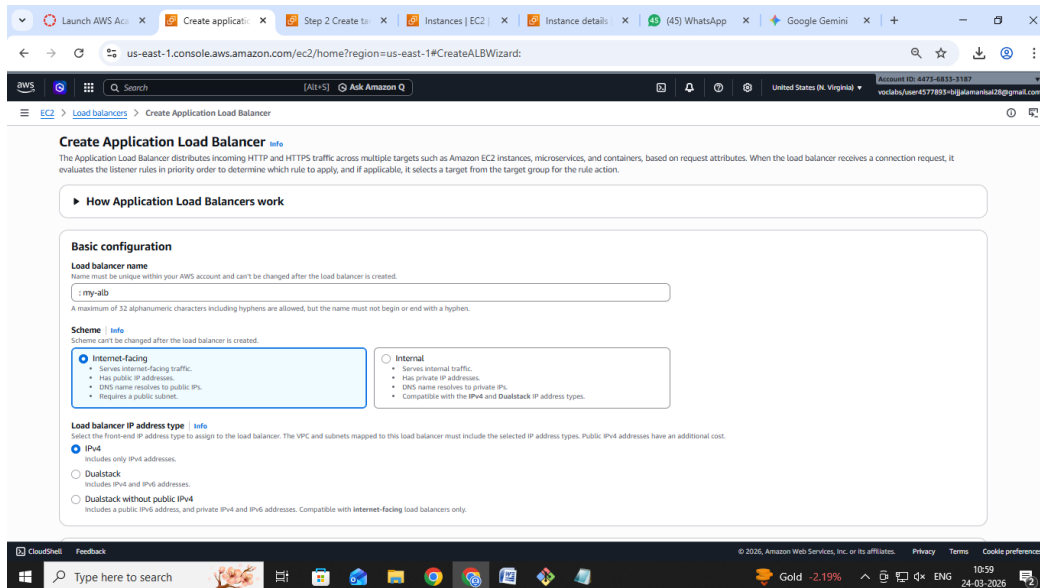
3. Listeners:

- Default: HTTP (80)



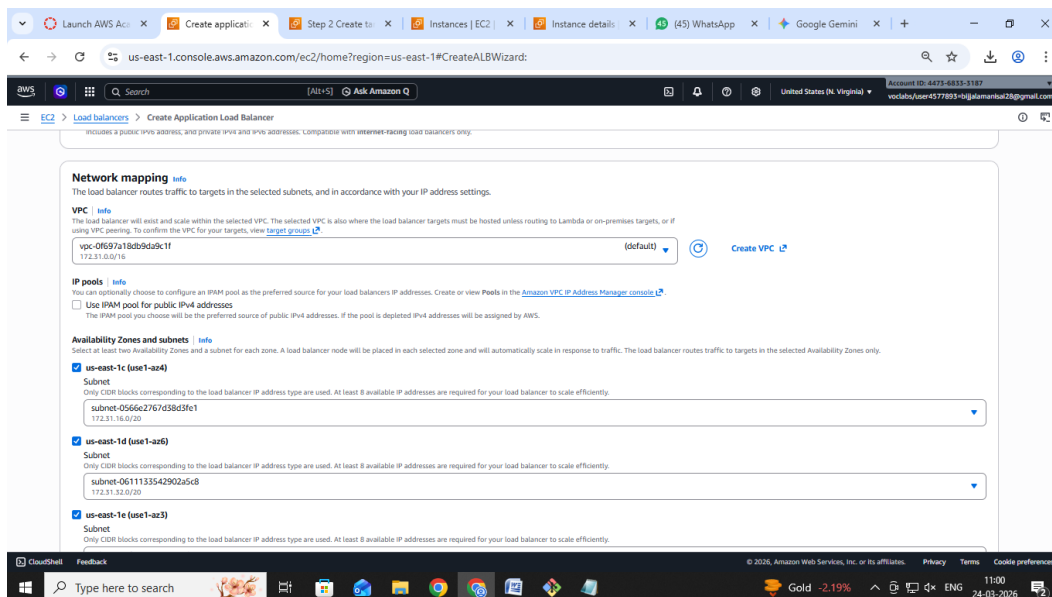
2.3 Configure Availability Zones

- Select the default **VPC** where your EC2 instances exist
- Select **two /all subnets** (one for each instance if possible)



2.4 Configure Security Groups

- Choose the **lb-sg** security group created earlier



2.5 Configure Target Groups

1. Create a new target group:

- Name: web-servers-tg
- Target type: Instance
- Protocol: HTTP
- Port: 80
- Health checks: / (default)

2. Register Targets:

- Select EC2 INSTANCES HERE ---**webserver-1** and **webserver-2**
- Click **Include as pending** → **Register**

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.*

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

- ☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or EC2 instances for automatic management.
Suitable for: **ALB** **NLB** **GWLB**
- ☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: **ALB** **NLB** **GWLB**
- ☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: **ALB**
- ☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: **NLB**

Target group name
Name must be unique per Region per AWS account.
web-servers-tg
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 14/32

Protocol
Protocol for communication between the load balancer and targets.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.
☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn](#)

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address
☒ i-09b7ac089ca8f5da	webserver-2	Running	launch-wizard-15	us-east-1c	172.31.31.119
☒ i-012de146d13ce9827	webserver-1	Running	launch-wizard-14	us-east-1c	172.31.31.71

0 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.
80
1-65535 (Separate multiple ports with commas)

[Include as pending below](#)

2 selections are now pending below. Include more or register targets when ready.

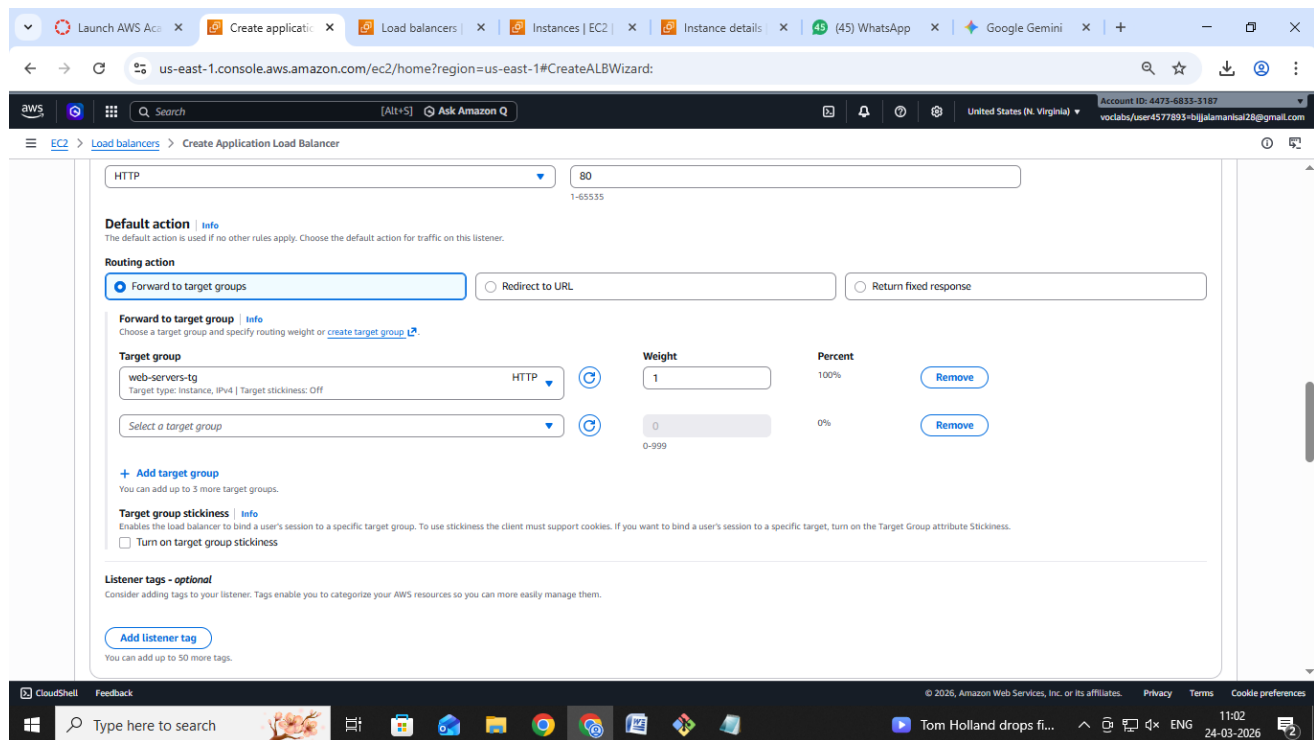
Review targets

Targets (2)

[Remove all pending](#)

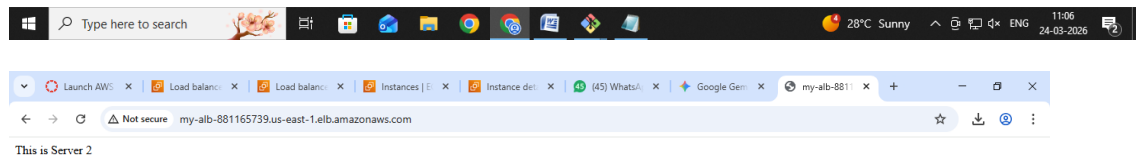
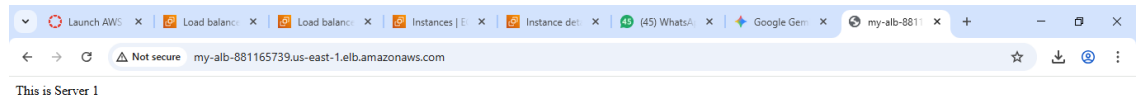
2.6 Review and Create

- Review all settings → Click **Create Load Balancer**
- AWS will take a few minutes to provision the ALB



Step 3: Verify Load Balancer

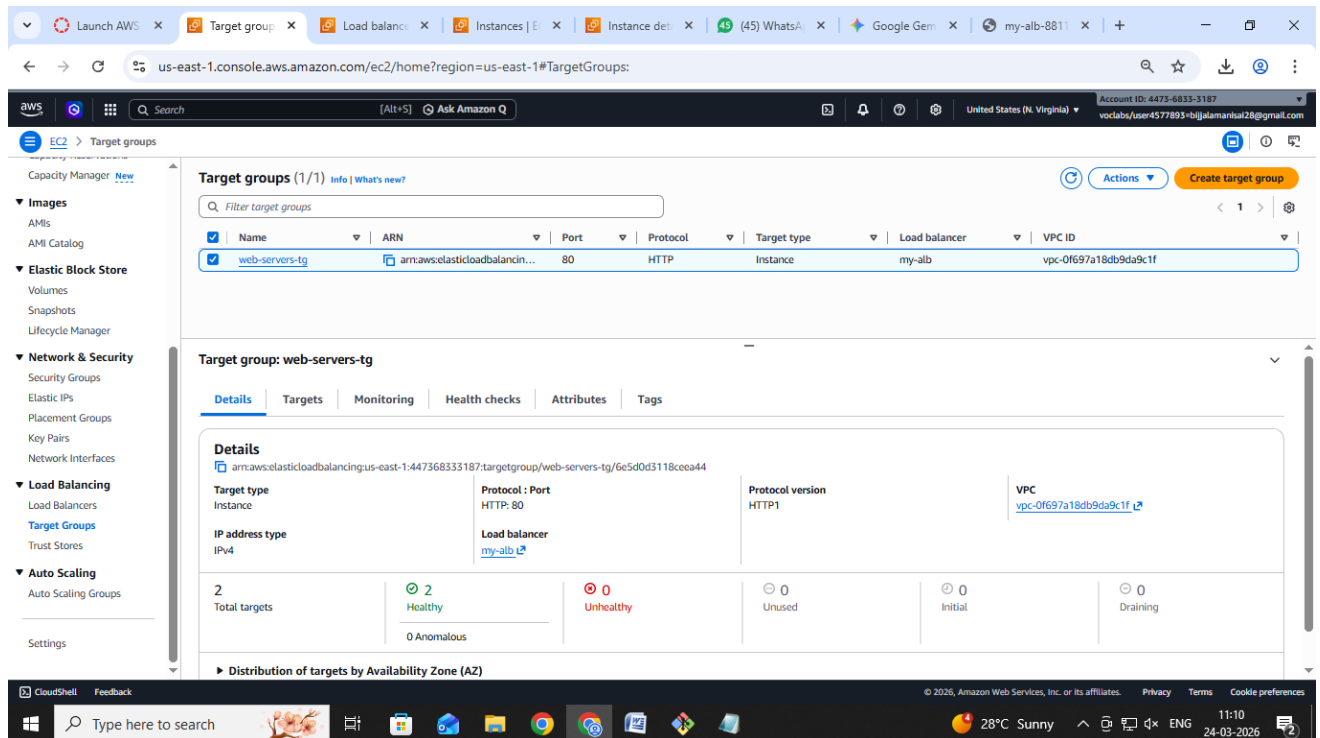
1. Go to **Load Balancers** → **Select ALB** → **Description** → **DNS name**
2. Copy the **DNS name** (something like my-alb-123456.us-east-1.elb.amazonaws.com)
3. Paste it in a browser → You should see either **WebServer-1** or **WebServer-2** page.
 - Refresh multiple times → Load balancer distributes traffic between the two instances



Step 4: Optional – Test Health Checks

- Go to **Target Groups** → **web-servers-tg** → **Targets**
- Ensure **Status** of both instances is healthy

If a server is unhealthy, check the bootstrapping script or security group rules.



Step-by-Step: Auto Scaling Group with Load Balancer

Step 1: Create an AMI from Existing EC2 Instance

1. Go to **EC2 Console** → **Instances**
2. Select the EC2 instance you already configured with bootstrapping
3. Click **Actions** → **Image and Templates** → **Create Image**
4. **Provide Image Name:** e.g., my-webserver-AMI
5. Optional: Add description
6. Click **Create Image**
7. Wait for the AMI to become **available** (status: available in **AMIs** section)

✓ This AMI will be used for launching new instances automatically in ASG.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-012de14e512ce9827

EC2 > Instances > i-012de14e512ce9827

Instance summary for i-012de14e512ce9827 (webserver-1) Info

Updated less than a minute ago

Instance ID
i-012de14e512ce9827

Public IPv4 address
52.204.205.155 | [open address](#)

Instance state
Running

Private IPv4 addresses
172.31.31.71

Public DNS
ec2-52-204-205-155.compute-1.amazonaws.com

Private IP DNS name (IPv4 only)
ip-172-31-31-71.ec2.internal

Instance type
t2.micro

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Hostname type
IP name: ip-172-31-31-71.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
52.204.205.155 [Public IP]

VPC ID
vpc-0f697a18db9da9c1f

Subnet ID
subnet-0566e2767d38d3fe1

Instance ARN
arn:aws:ec2:us-east-1:447368333187:instance/i-012de14e512ce9827

IAM role
-

IMDSv2
Required

Operator
-

Actions
Connect, Instance state, Instance diagnostics, Instance settings, Networking, Security, Image and templates, Storage, Monitor and troubleshoot

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance details Info

CloudShell Feedback

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28°C Sunny 11:13 24-03-2026

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateImage:instanceId=i-012de14e512ce9827

EC2 > Instances > i-012de14e512ce9827 > Create image

Create image Info

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 Instance. You can create an image from the configuration of an existing instance.

Image details

Instance ID
i-012de14e512ce9827 (webserver-1)

Image name
my-webserver-AMI

Image description - optional
Image description

☒ **Reboot instance**
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
EBS	/dev/x...	Create new snapshot from v...	8	EBS General Purpose SSD - ...	3000		☒ Enable	☐ Enable

[Add volume](#)

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

CloudShell Feedback

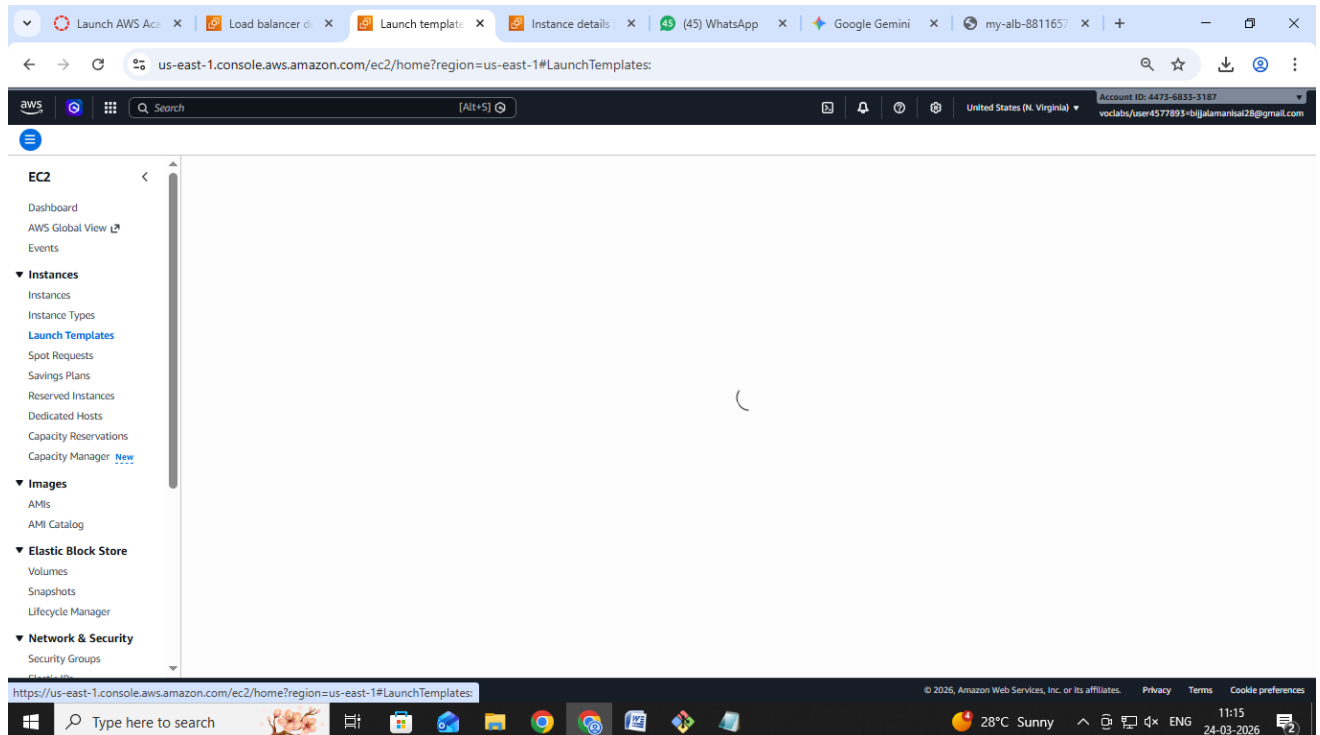
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28°C Sunny 11:14 24-03-2026

Step 2: Create a Launch Template

1. Go to **EC2** → **instances->Launch Templates** → **Create Launch Template**
2. **Launch Template Name:** my-launch-template
3. **AMI:** Select the AMI created in Step 1 (my-webserver-AMI)
4. **Instance Type:** t2.micro (or your choice)
5. **Key Pair:** Select your existing key pair
6. **Security Groups:** Select the **Load Balancer Security Group** (allow HTTP 80)
7. **User Data:** Optional (already baked into AMI)
8. Click **Create Launch Template**

✓ **Launch Template is ready for Auto Scaling.**



Launch AWS Academy x Create launch template x Instance details | EC2 x (45) WhatsApp x Google Gemini x my-alb-881165739.us x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

Search [Alt+S]

United States (N. Virginia) Account ID: 4473-6833-3187 voclabs/user4577893-bijalamanisai28@gmail.com

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '"', '@'.

Template version description

Max 255 chars

Auto Scaling guidance | Info

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) | Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

▼ Summary

Software Image (AMI)
my-webserver-AMI
ami-0d7d9651631797eb1

Virtual server type (instance type)
t2.micro

Firewall (security group)
lb-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

Launch AWS Academy x Create launch template x Instance details | EC2 x (45) WhatsApp x Google Gemini x my-alb-881165739.us x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

Search [Alt+S]

United States (N. Virginia) Account ID: 4473-6833-3187 voclabs/user4577893-bijalamanisai28@gmail.com

Create launch template

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start

☐ Don't include in launch template ☒ Owned by me ☐ Shared with me

Amazon Machine Image (AMI)

my-webserver-AMI
ami-0d7d9651631797eb1
2026-03-24T05:44:38.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred

Description

-

Architecture x86_64 **AMI ID** ami-0d7d9651631797eb1

▼ Instance type | Info | Get advice

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

☒ All generations

[Compare instance types](#)

▼ Summary

Software Image (AMI)
my-webserver-AMI
ami-0d7d9651631797eb1

Virtual server type (instance type)
t2.micro

Firewall (security group)
lb-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

Launch AWS Academy x>Create launch template x>Instance details | EC2 x>(45) WhatsApp xGoogle Gemini xmy-alb-881165739.us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

Search[Alt+S]

United States (N. Virginia)Account ID: 4473-6833-3187vociabo/user4577893-bijalamanisai28@gmail.com

Search results

▼ Instance typeInfo | Get adviceAdvanced

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

Additional costs apply for AMIs with pre-installed software

All generations
Compare instance types

▼ Key pair (login)Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

week9

Create new key pair

▼ Network settingsInfo

SubnetInfo

Don't include in launch template

When you specify a subnet, a network interface is automatically added to your template.

Create new subnet

Availability ZoneInfo

Don't include in launch template

Enable additional zones

Firewall (security groups)Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group

Create security group

▼ Summary

Software Image (AMI)
my-webserver-AMI
ami-0d7d9651631797eb1

Virtual server type (instance type)
t2.micro

Firewall (security group)
lb-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Create launch template

CloudShellFeedback

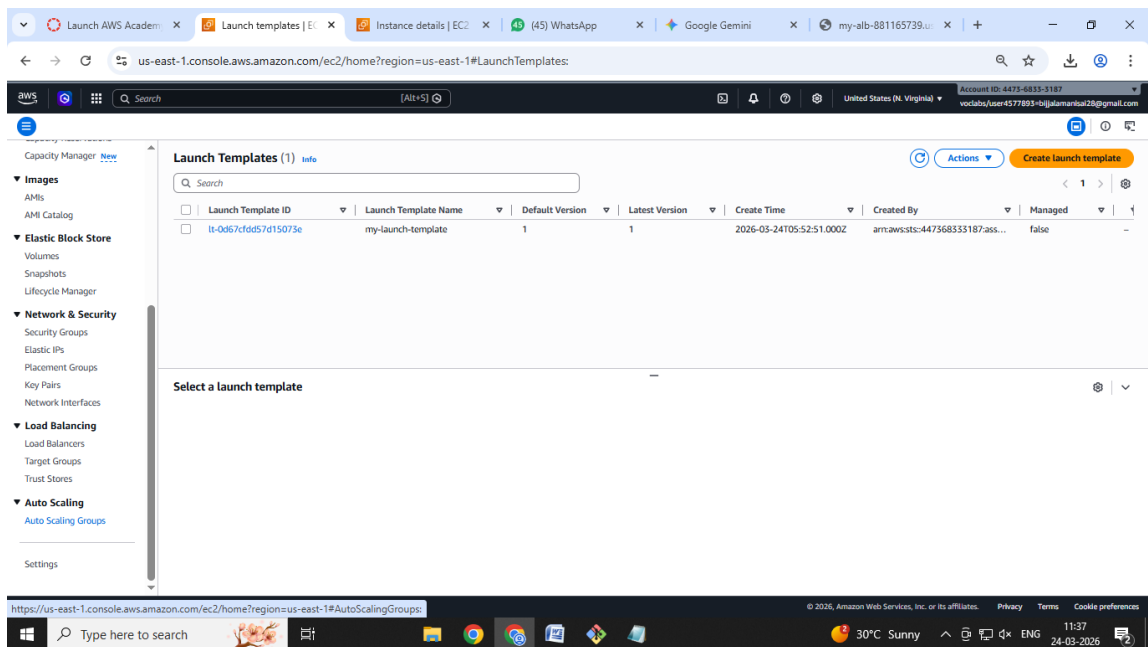
Type here to search

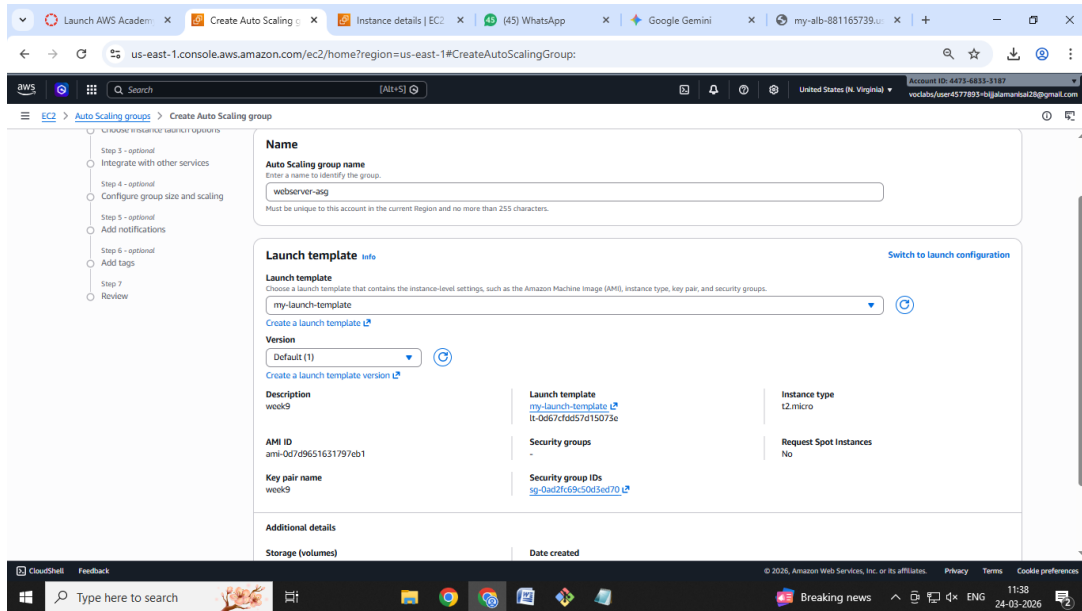
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28°C Sunny 11:22 24-03-2026

Step 3: Create Auto Scaling Group (ASG)

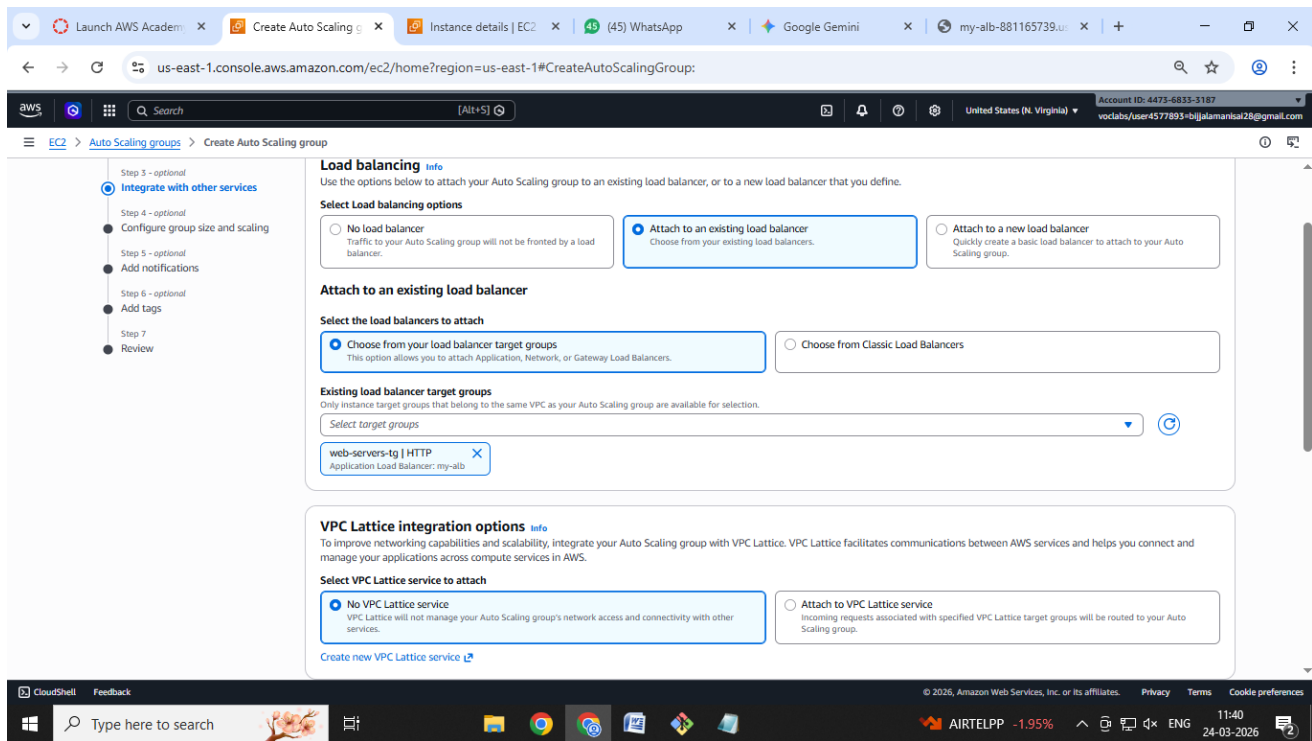
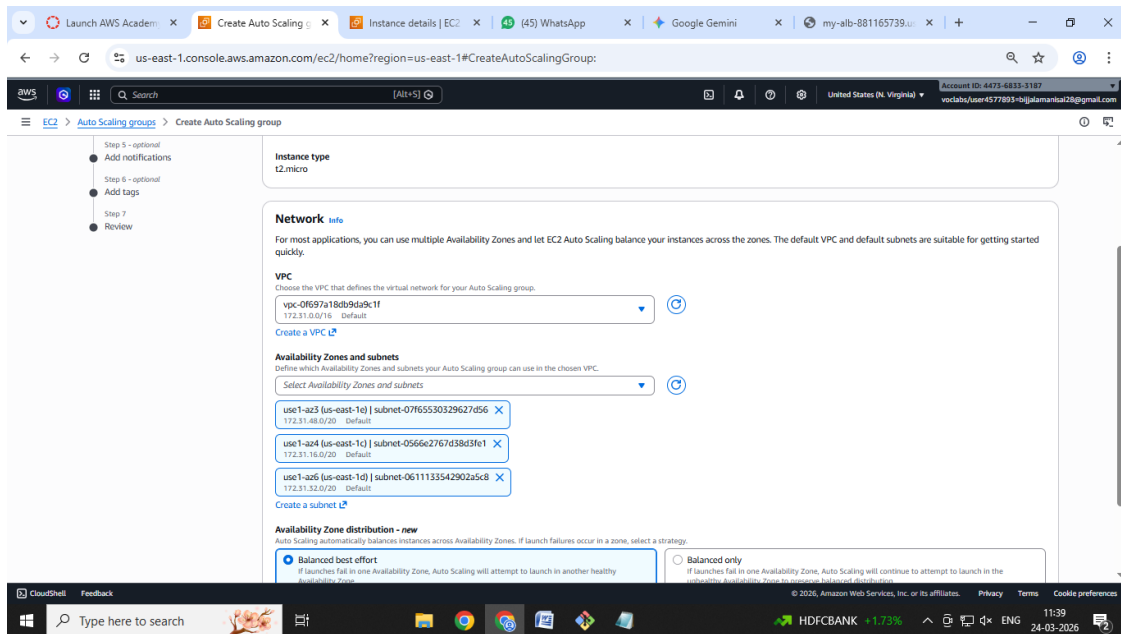
1. Go to **EC2** → **Auto Scaling Groups** → **Create Auto Scaling Group**
2. **Name:** webserver-asg
3. **Launch Template:** Select my-launch-template
4. **Template Version:** Select **Version 1**
5. Click **Next**





Step 3.1: Network Settings

1. **VPC:** Default VPC (or your existing VPC)
2. **Availability Zones (AZs):** Select **all** subnets (for high availability)
3. **Load Balancer:** Enable **Attach to an existing load balancer**
 - Select your **existing Load Balancer**
4. **Health Check:** HTTP
 - Set **Health Check Grace Period:** 300 seconds
5. Click **Next**



Step 3.2: Configure Group Size

1. **Desired Capacity:** 2 (initial instances)

2. **Minimum Capacity:** 1
3. **Maximum Capacity:** 4
4. Click **Next**

Step 1: Choose launch template or configuration

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: **Configure group size and scaling**

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Configure group size and scaling - optional [info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [info](#)
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling [info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
1

Max desired capacity
4

Equal or less than desired capacity

Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy [info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ **No scaling policies**
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ **Target tracking scaling policy**
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Step 3.3: Review and Create ASG

1. Review all settings
2. Click **Create Auto Scaling Group**

✓ Auto Scaling Group is now ready and attached to your Load Balancer.

Step 4: Verify Load Balancer and Auto Scaling

1. Go to **Load Balancers** → **Select your Load Balancer**
2. Click **Instances** → **Target Group** → **Registered Targets**
 - You will see **2 healthy instances** initially

Register targets

Select instances, specify ports, and add the instances to the list of pending targets. Repeat to add additional combinations of instances and ports to the list of pending targets. Once you are satisfied with your selections, click Register pending targets.

Available instances (2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-09b7ac089eaa8f5da	webserver-2	Running	launch-wizard-15	us-east-1c	172.31.31.119	subnet-0566c276d38d3fe1
i-012de14e512ce9827	webserver-1	Running	launch-wizard-14	us-east-1c	172.31.31.71	subnet-0566c276d38d3fe1

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with comma)

Include as pending below

Review targets

Targets (2)

Remove all pending

3. Check Auto Scaling Group → Instances

- You may see up to 4 instances if scaling triggers

Instances (1/4)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
webserver-1	i-012de14e512ce9827	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-52-204-205-155.co...	52.204.205.155
webserver-2	i-09b7ac089eaa8f5da	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-44-222-209-8.com...	44.222.209.8
	i-0958baa330ca5d128	Running	t2.micro	Initializing	View alarms +	us-east-1c	ec2-18-234-212-241.co...	18.234.212.241

i-012de14e512ce9827 (webserver-1)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

Instance ID: i-012de14e512ce9827

Public IPv4 address: 52.204.205.155 | open address

Instance state: Running

Public DNS: ec2-52-204-205-155.compute-1.amazonaws.com | open address

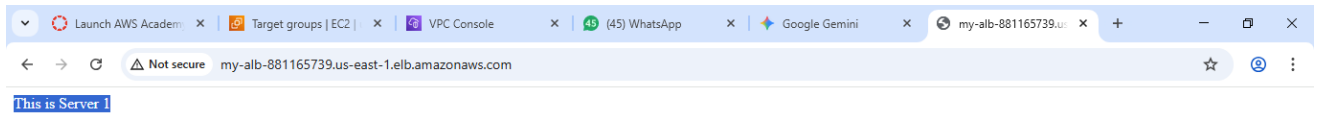
4. Test auto recovery:

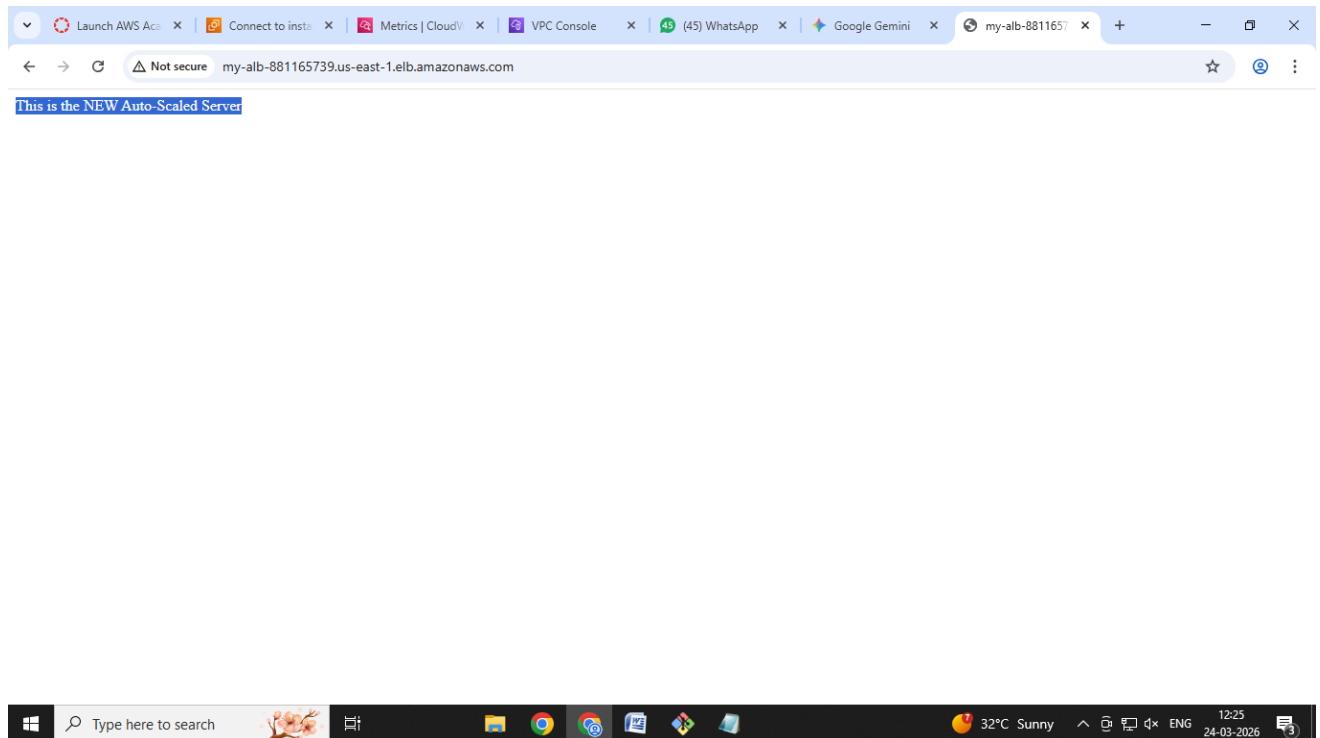
- **Terminate one of the instances**
- After a few seconds, **ASG will launch a new instance automatically**
- Traffic will automatically be **distributed by the Load Balancer**

Step 5: Test the Load Balancer

1. Copy the **Load Balancer DNS Name** from ALB
2. Open in a browser → You should see your web page from one of the instances
3. Refresh multiple times → traffic alternates between instances

✓ This confirms **Auto Scaling + Load Balancer** is working correctly.





Summary of Flow

1. **Create AMI** → snapshot of working EC2
 2. **Create Launch Template** → using AMI, security group
 3. **Create Auto Scaling Group** → attach Launch Template, set min/desired/max, attach Load Balancer
 4. **Verify instances** → ASG launches/replaces instances automatically
 5. **Test Load Balancer** → distributes traffic between all active instances
-

Scenario Based Questions (ELB & Auto Scaling)

1) A company hosts a web application on two EC2 instances... Which AWS service can distribute incoming traffic evenly across both servers?

Answer: Elastic Load Balancer (ELB)

2) A developer created multiple EC2 instances... single entry point instead of multiple IPs?

Answer: Elastic Load Balancer (ELB)

3) Application receives heavy traffic during peak hours... automatically increase/decrease servers?

Answer: Auto Scaling (Auto Scaling Group)

4) Traffic is being sent to an unhealthy server... which feature stops this?

Answer: Health Checks (ELB Health Checks)

5) Launch EC2 instances when CPU > 70%?

Answer: Auto Scaling Group with Scaling Policy (Target Tracking / CloudWatch Alarm-based scaling)

6) Application receives global traffic... how can ELB help?

Answer: Elastic Load Balancer distributes incoming traffic across multiple EC2 instances across Availability Zones to ensure high availability and fault tolerance.

7) Auto Scaling Group not launching instances... what might be missing?

Answer: Missing or incorrect Scaling Policy / Launch Template / Maximum Capacity setting

8) Maintain minimum 2 servers and scale when needed?

Answer: Define:

- Minimum Capacity = 2
- Desired Capacity = 2
- Maximum Capacity = higher value (e.g., 5 or more)

9) App crashes during traffic spikes... how does ELB + Auto Scaling help?

Answer:

- ELB distributes traffic evenly

- Auto Scaling adds/removes instances automatically

→ prevents overload and improves reliability

10) Auto Scaling Group not attached to Load Balancer... what problem?

Answer: New instances won't receive traffic, leading to uneven load and wasted resources

11) Replace unhealthy EC2 instances automatically?

Answer: Auto Scaling Group (with Health Checks and Instance Replacement)

12) Monitor EC2 performance and trigger scaling?

Answer: Amazon CloudWatch

13) Handle sudden traffic spikes during promotion?

Answer: Auto Scaling automatically adds more EC2 instances during high traffic and removes them when demand decreases

14) Route traffic based on HTTP/HTTPS protocols?

Answer: Application Load Balancer (ALB)

15) Highly available architecture with load balancing + scaling?

Answer:

- Elastic Load Balancer (ELB)
- Auto Scaling Group (ASG)